
PROBLEM DEFINITION DOCUMENT

Diabetes Risk & Patient Health Analysis

1. Problem Overview

Diabetes mellitus is a long-term metabolic dysfunction where the body fails to control blood glucose levels. The International Diabetes Federation (2021) reports that more than 537 million adults worldwide are living with diabetes, and this number is projected to grow to 783 million by 2045.

In Sri Lanka, diabetes is one of the most common non-communicable illnesses, with a prevalence rate of 10.3% among adults, making it a major challenge in the national healthcare system (Katulanda et al., 2012). Almost half of all diabetics globally remain undiagnosed.

2. Problem Statement

This project addresses the early detection of diabetes mellitus using machine learning classification techniques. Conventional diagnostic processes can be slow, expensive, and inaccessible in resource-constrained environments. Machine learning offers an encouraging alternative by rapidly identifying at-risk individuals at low cost, based on routine clinical measurements.

The project aims to develop a binary classification model that predicts the presence of diabetes in a patient based on 8 diagnostic health measures. The target outcome is binary: 1 (diabetic) or 0 (non-diabetic).

ions and Medical Care, pp. 261–265.